

# Brown phases in game semantics: a canonical two-bit split and a divisibility obstruction

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2026-08-09

## Abstract

The Brown invariant of a nonsingular  $\mathbb{Z}/4$ -valued quadratic form is the eighth-root phase of a four-class Gauss census. We determine exactly what a game-theoretic reading of that census adds to the characteristic-two Arf problem. Every Brown form has a canonical decomposition

$$q = \widehat{\ell} + 2Q,$$

where  $\ell = q \bmod 2$  is linear and  $Q$  is an ordinary  $\mathbb{F}_2$ -quadratic form. Consequently the four residue classes are precisely the joint census of one linear game bit and the ordinary quadratic bit already isolated by the **tis** problem; the Brown phase introduces no second nonlinear game-semantic problem. We also prove a complementary global no-go: an ambient-coherent Brown refinement of the additive group of all short games is identically zero, because that group is 2-divisible. Finally, the cyclic extension  $C_4 \rightarrow C_8 \rightarrow C_2$  shows that a bare central extension by  $C_4$  does not determine a Brown form: a section or phase framing is essential. Together these results dispose of the independent Brown invariant/output question under an explicit synchronized-charge contract, without claiming a single-game outcome construction or a solution of **tis** itself.

## 1 The exact question

A Brown form on an  $\mathbb{F}_2$ -space  $V$  is a function

$$q : V \longrightarrow \mathbb{Z}/4$$

with  $q(0) = 0$  and

$$q(x + y) = q(x) + q(y) + 2b(x, y), \tag{1}$$

where  $b : V \times V \rightarrow \mathbb{F}_2$  is symmetric and bilinear. Substituting  $y = x$  shows

$$b(x, x) = q(x) \bmod 2. \tag{2}$$

Thus  $b$  need not be alternating. When  $b$  is nonsingular, the classical Brown invariant  $\beta(q) \in \mathbb{Z}/8$  is defined by the Gauss sum

$$\sum_{x \in V} i^{q(x)} = 2^{\dim(V)/2} \zeta_8^{\beta(q)}. \tag{3}$$

This is standard mathematics due to Brown and Wall [1, 3]; the repository's **brown\_f2** implementation also records the singular radical boundary.

The phrase “game-theoretic reading” has three logically different senses:

- (i) a four-class charge or outcome attached to positions in one selected finite  $\mathbb{F}_2$ -space;
- (ii) a quadratic colour attached intrinsically to short-game *values* and compatible with disjunctive sum and ambient enlargement; or
- (iii) a framed evaluator which is handed  $q$  and reports its residue.

Without separating these, four raw labels prove nothing: partizan normal play, normal/misere pairs, loopy starter pairs, and scoring games all provide at least four labels. The results below reduce (i) algebraically to a synchronized linear/quadratic charge pair, rule out (ii) nontrivially, and identify (iii) as tautological rather than a new bridge.

## 2 The canonical two-bit decomposition

Write  $\rho : \mathbb{Z}/4 \rightarrow \mathbb{F}_2$  for reduction modulo two and  $\widehat{\cdot} : \mathbb{F}_2 \rightarrow \mathbb{Z}/4$  for the distinguished set-theoretic section  $0 \mapsto 0, 1 \mapsto 1$ . It obeys

$$\widehat{a + c} = \widehat{a} + \widehat{c} + 2ac \quad (a, c \in \mathbb{F}_2), \quad (4)$$

where the left addition is in  $\mathbb{F}_2$  and the right addition is in  $\mathbb{Z}/4$ .

**Theorem 1** (canonical Brown split). **[PROVED]** *Let  $(q, b)$  satisfy (1). There are unique maps*

$$\ell : V \rightarrow \mathbb{F}_2, \quad Q : V \rightarrow \mathbb{F}_2$$

*such that*

$$q(x) = \widehat{\ell(x)} + 2Q(x). \quad (5)$$

*They have the following properties.*

- (i)  $\ell = \rho \circ q$  is linear.
- (ii)  $Q$  is an ordinary characteristic-two quadratic form with alternating polar form

$$B_Q(x, y) = b(x, y) + \ell(x)\ell(y). \quad (6)$$

- (iii) *Conversely, every linear  $\ell$  and every ordinary  $\mathbb{F}_2$ -quadratic form  $Q$  determine a unique Brown form by (5); its symmetric polar form is*

$$b = B_Q + \ell \otimes \ell.$$

*Hence Brown forms are canonically equivalent to pairs  $(\ell, Q)$ .*

*Proof.* Reducing (1) modulo two kills  $2b$ , so  $\ell = \rho q$  is additive and therefore  $\mathbb{F}_2$ -linear. Each residue  $q(x)$  has a unique expression (5), which defines  $Q$  and proves uniqueness.

Apply (4) to  $\ell(x + y) = \ell(x) + \ell(y)$  and compare the two expressions for  $q(x + y)$ . Cancelling the distinguished lifts and dividing the remaining even equality by two gives

$$Q(x + y) = Q(x) + Q(y) + b(x, y) + \ell(x)\ell(y).$$

Thus the displayed  $B_Q$  is the polar form of  $Q$ . Equation (2) and  $\ell(x)^2 = \ell(x)$  give

$$B_Q(x, x) = b(x, x) + \ell(x)^2 = \ell(x) + \ell(x) = 0,$$

so it is alternating; bilinearity and symmetry follow from those of  $b$  and  $\ell \otimes \ell$ .

Conversely, let  $\ell$  be linear and  $Q$  have alternating polar  $B_Q$ . Set  $q = \widehat{\ell} + 2Q$  and  $b = B_Q + \ell \otimes \ell$ . Equation (4) and the polarization law for  $Q$  give (1); moreover  $b(x, x) = \ell(x) = q(x) \bmod 2$ . The two constructions are inverse by the uniqueness of the two binary digits of every element of  $\mathbb{Z}/4$ .  $\square$

**Corollary 2** (the four-class census is a joint binary census). **[PROVED]** *Under Theorem 1, the four Brown classes are*

| $q(x)$ | $\ell(x)$ | $Q(x)$ | $i^{q(x)}$ |
|--------|-----------|--------|------------|
| 0      | 0         | 0      | +1         |
| 1      | 1         | 0      | +i         |
| 2      | 0         | 1      | -1         |
| 3      | 1         | 1      | -i         |

*In particular*

$$\sum_x i^{q(x)} = \sum_x i^{\ell(x)} (-1)^{Q(x)}. \quad (7)$$

*If  $W(R) = \sum_x (-1)^{R(x)}$ , then more explicitly*

$$\sum_x i^{q(x)} = \frac{1+i}{2} W(Q) + \frac{1-i}{2} W(Q + \ell). \quad (8)$$

*When the full Gauss sum is nonzero—in particular when  $b$  is nonsingular— $\beta$  is the octant of the joint  $(\ell, Q)$  imbalance, or equivalently of the correlated pair of ordinary quadratic signed censuses  $Q$  and  $Q + \ell$ . It is not generally the product of a marginal linear census and the Arf bias of  $Q$ . On the doubled nonsingular slice  $\ell = 0$ , the two Walsh sums agree and this reduces to the shipped identity  $\beta(2Q) = 4 \operatorname{Arf}(Q)$ . For singular forms the same census identity holds, while the phase belongs to the nonsingular core and the radical data is recorded separately.*

*Proof.* The table and (7) follow immediately from  $q = \widehat{\ell} + 2Q$ . The identity

$$i^\ell = \frac{1+i}{2} + \frac{1-i}{2} (-1)^\ell$$

gives (8). For  $\ell = 0$ , the Gauss sum is the ordinary real quadratic Gauss sum, whose sign is  $(-1)^{\operatorname{Arf}(Q)}$ .  $\square$

## 2.1 The Brown phase in ordinary quadratic coordinates

The split gives more than a four-colour relabelling: it computes every Brown octant from ordinary characteristic-two data.

**Theorem 3** (phase formulas). **[PROVED]** *Suppose the symmetric Brown polar form  $b$  is nondegenerate and put  $n = \dim V$ .*

(i) *If  $n$  is even,  $B_Q$  is nondegenerate. Let  $a$  be the unique vector satisfying  $\ell(x) = B_Q(a, x)$  for every  $x$ . Then*

$$\beta(q) = 4 \operatorname{Arf}(Q) + 2Q(a) \pmod{8}. \quad (9)$$

(ii) *If  $n$  is odd,  $\operatorname{rad} B_Q = \langle w \rangle$  is one-dimensional,  $\ell(w) = 1$ , and  $B_Q$  restricts nondegenerately to  $H = \ker \ell$ . Then*

$$\beta(q) = 4 \operatorname{Arf}(Q|_H) + \begin{cases} 1, & Q(w) = 0, \\ 7, & Q(w) = 1 \end{cases} \pmod{8}. \quad (10)$$

*Proof.* For a nonsingular symmetric form over  $\mathbb{F}_2$ , the characteristic functional  $x \mapsto b(x, x)$  has a unique representing vector  $c$ . The standard orthogonal line/hyperbolic-plane decomposition shows  $b(c, c) = n \bmod 2$ . Since  $\ell(x) = b(c, x)$  and  $B_Q = b + \ell \otimes \ell$ , its kernel equation is

$$B_Q(v, -) = 0 \iff v = \ell(v)c.$$

Thus  $B_Q$  is nonsingular when  $n$  is even; when  $n$  is odd its radical is  $\langle c \rangle$ , with  $\ell(c) = 1$ , and its restriction to  $H = \ker \ell$  is nonsingular. Put  $w = c$  in the odd case.

In even dimension, nondegeneracy supplies the unique  $a$  with  $B_Q(a, -) = \ell$ . Translation by  $a$  gives

$$Q(x + a) = Q(x) + Q(a) + \ell(x),$$

and hence  $W(Q + \ell) = (-1)^{Q(a)}W(Q)$ . The ordinary quadratic Gauss formula gives  $W(Q) = (-1)^{\text{Arf}(Q)}2^{n/2}$ . Substitution in (8) yields (9).

In odd dimension, pair the two cosets as  $h$  and  $h + w$ ,  $h \in H$ . Since  $w \in \text{rad } B_Q$ ,

$$\sum_{x \in V} i^{q(x)} = \sum_{h \in H} (-1)^{Q(h)} (1 + i(-1)^{Q(w)}).$$

The first factor is  $(-1)^{\text{Arf}(Q|_H)}2^{(n-1)/2}$ ; the second has phase  $\zeta_8$  or  $\zeta_8^7$ . This proves (10).  $\square$

*Remark 4* (degenerate forms). Let  $R = \text{rad } b$ . Equation (2) gives  $\ell|_R = 0$ , hence  $q|_R = 2Q|_R$ . If  $Q|_R \neq 0$ , translation by a vector with  $Q = 1$  cancels the full Gauss sum. If  $Q|_R = 0$ , the pair  $(\ell, Q)$  descends to  $V/R$  and Theorem 3 applies to the nonsingular quotient. This is exactly the repository's `radical_anisotropic` boundary.

*Remark 5* (the split is twisted under addition). Theorem 1 is a canonical bijection of data, not a direct-product law. Under the Brown–Wall orthogonal sum, if  $q_1, q_2$  have coordinates  $(\ell_1, Q_1)$  on  $V_1$  and  $(\ell_2, Q_2)$  on  $V_2$ , then on  $V_1 \oplus V_2$

$$\begin{aligned} \ell(x_1, x_2) &= \ell_1(x_1) + \ell_2(x_2), \\ Q(x_1, x_2) &= Q_1(x_1) + Q_2(x_2) + \ell_1(x_1)\ell_2(x_2). \end{aligned} \tag{11}$$

The same formula holds for pointwise addition on one source whenever that sum is considered. The last term is the binary carry from (4): it prevents this coordinate split from being a direct-product law and reflects the cited Brown–Wall  $\mathbb{Z}/8$  Witt structure; it is not by itself a proof of that classification. A positionwise joint readout retains the carry and the correlation; separate marginal counts do not.

### 3 What the split means for play

**Definition 6** (charge readout). A binary or four-class *charge readout* is a game rule whose complete play assigns a terminal element of  $\mathbb{F}_2$  or  $\mathbb{Z}/4$ . It is weaker than a normal-play  $P/N$  realization: the terminal charge may be strategically forced without itself being the winner class.

**Definition 7** (synchronized four-charge contract). Two charge readouts on the same input may be observed jointly and their ordered pair may be relabelled by a fixed bijection. Equivalence in this contract means equality of the resulting terminal labels for every input and complete play. It does *not* assert that the two game trees factor, that their strategies are independent, or that the pair is already one of the canonical outcome classes of a single normal-, misere-, or loopy-play game.

**Proposition 8** (no independent over charge). **[PROVED]** *A four-class charge readout realizing a Brown form  $q$  is equivalent, by deterministic relabelling, to the joint readout  $(\ell, Q)$  of Theorem 1. Conversely, joint readouts of  $\ell$  and  $Q$  give a four-class readout by  $(a, c) \mapsto \hat{a} + 2c$ , provided the semantics admits synchronized products of two charge channels on the same input.*

*Once  $\ell$  is supplied, its coordinate is a solved local linear charge layer: after choosing any basis, attach a  $*$  component to precisely those active coordinates on which  $\ell$  is one; their disjunctive sum has Grundy value  $*\ell(x)$ . All nonlinear dependence of the terminal charge on the input is therefore the ordinary quadratic readout  $Q$ . If one strengthens “charge readout” to “ $P$  exactly when  $Q = 0$ ”, that remaining requirement is a generalized **tis**-type problem; it is literally the existing Gold-form **tis** problem on the doubled slice  $\ell = 0$ ,  $q = 2Q$ .*

*Proof.* The residue-to-bit map is the table in Corollary 2, and its inverse is  $(a, c) \mapsto \hat{a} + 2c$ ; this is an observational equivalence of terminal labels, not a factorization theorem for game trees. For the linear realization, write  $x = \sum_i x_i e_i$ . The sum of the tagged  $*$  components has Grundy coordinate  $\sum_i x_i \ell(e_i) = \ell(x)$ . Changing basis changes the presentation, not the resulting linear functional. The final sentence is the definition of the **tis** target applied to the ordinary quadratic form  $Q$ .  $\square$

*Remark 9* (the exact residual boundary). **[BOUNDARY]** Proposition 8 does not turn a forced terminal charge into a normal-play winner. The repository’s  $\sigma$ -valued echo-FIFO construction is therefore genuine progress toward the  $Q$  bit but does not, without the general linking theorem and an outcome recasting, solve **tis**. Nor is a synchronized ordered pair automatically one canonical normal-, misere-, or loopy-play outcome of a single game; product closure is part of the four-charge contract. The point is narrower and complete: Brown’s four classes add a linear axis and a correlated second census, not a new algebraic kind of nonlinear charge/output problem.

## 4 The global value-level obstruction

The previous section concerns a selected finite  $\mathbb{F}_2$ -space. A stronger naturality demand is that  $q$  belong to short-game values themselves rather than to one root-incomplete slice.

**Definition 10** (ambient-coherent Brown value datum). For every finitely generated subgroup  $M \leq \text{ShUg}$ , choose

$$q_M : M \rightarrow \mathbb{Z}/4, \quad b_M : M \times M \rightarrow \mathbb{F}_2$$

satisfying (1), with  $b_M$  biadditive. The family is *ambient-coherent* if restriction along every inclusion  $M \subseteq N$  preserves both functions. This is a generalized Brown-law datum on arbitrary game groups, deliberately stronger than a classical Brown form whose domain is elementary abelian.

**Theorem 11** (short-game Brown collapse). **[PROVED]** *Every ambient-coherent Brown value datum on short games is zero:*

$$b_M = 0, \quad q_M = 0$$

*for every finitely generated  $M \leq \text{ShUg}$ .*

*Proof.* Moews proves

$$\text{ShUg} \cong \bigoplus_{\mathbb{N}} \mathbb{Z}[1/2] \oplus \bigoplus_{\mathbb{N}} (\mathbb{Z}[1/2]/\mathbb{Z}), \quad (12)$$

so multiplication by two, and hence by four, is surjective [2, Theorem 9].

Fix  $x, y \in M$ . Choose  $u \in \text{ShUg}$  with  $2u = x$  and enlarge  $M$  to the finitely generated subgroup  $N = \langle M, u \rangle$ . Coherence and bilinearity give

$$b_M(x, y) = b_N(2u, y) = 2b_N(u, y) = 0 \quad \text{in } \mathbb{F}_2.$$

Thus all polar corrections vanish and every  $q_M$  is additive. Choose  $v \in \text{ShUg}$  with  $4v = x$  and put  $P = \langle M, v \rangle$ . Then

$$q_M(x) = q_P(x) = q_P(4v) = 4q_P(v) = 0 \quad \text{in } \mathbb{Z}/4.$$

□

**Corollary 12** (no exponent-two Brown quotient). **[PROVED]** *Every additive map from  $\text{ShUg}$  to an exponent-two group is zero; in particular  $\text{ShUg}/2\text{ShUg} = 0$ . Hence no nonzero  $\mathbb{F}_2$  Brown module can be an additive quotient or homomorphic image of the full short-game group. This does not kill the intrinsic subgroup  $\text{ShUg}[2]$ : it contains  $*$  and its root-incomplete finite subspaces support nonzero local Brown forms.*

*Proof.* If  $f : \text{ShUg} \rightarrow A$  is additive,  $2A = 0$ , and  $x = 2u$ , then  $f(x) = 2f(u) = 0$ . □

This theorem is stronger than the earlier game-exterior obstruction on this specific target. It needs no Clifford algebra, coefficient embedding, or torsion hypothesis: the Brown polarization law and ambient roots suffice. The kernel-checked file `formal/0gdoad/BrownGame.lean` proves the abstract two-divisible/exponent-four implication and the  $\mathbb{Z}/4$  specialization; Equation (12) remains an explicit source-pinned input rather than a Lean axiom.

## 5 Why a bare central extension does not supply the phase

The characteristic-two extraspecial construction does not lift verbatim from  $C_2$  to  $C_4$ . The obstruction is not commutativity but missing section data.

**Proposition 13** (minimal framed cyclic model). **[PROVED]** *Consider the abelian exact sequence*

$$0 \longrightarrow C_4 \xrightarrow{a \mapsto 2a} C_8 \longrightarrow C_2 \longrightarrow 0.$$

*Choosing  $1 \in C_8$  as the lift of the nonzero quotient element gives the odd Brown line  $q(1) = 1$ ; choosing  $3 \in C_8$  gives  $q(1) = 3$ . Both have  $b(1, 1) = 1$  and Brown phases 1 and 7, respectively. Therefore the bare extension determines neither  $q$  nor  $\beta$ .*

*Proof.* Let  $z = 2 \in C_8$  generate the embedded  $C_4$ . For the first section,  $2 \cdot 1 = z^1$  in additive notation, while for the second,  $2 \cdot 3 = 6 = z^3$ . In either case, substituting the nonzero vector twice in (1) forces  $b(1, 1) = 1$ . The Gauss sums are  $1+i$  and  $1-i$ , hence the stated phases. More generally, changing a lift  $g$  to  $z + g$  changes its double by  $2z$ ; the square exponent is not fibre-invariant. □

The finite model and the non-fibre-invariance statement are also kernel-checked in `BrownGame.lean`. It corrects the tempting but false claim that a  $C_4$ -central extension “corresponds exactly” to a Brown form. What corresponds to the full refinement is a suitably section-framed extension. Its commutator remembers only alternating information; the admissible section/framing carries the missing quadratic lift data, including the diagonal phase.

## 6 Disposition of over

The three readings from Section 1 now have exact answers.

- (1) **[PROVED]** On a selected finite  $\mathbb{F}_2$ -space, the Brown four-class census is canonically the joint census of a locally realizable linear charge  $\ell$  (once supplied) and one ordinary quadratic bit  $Q$ . Its complex imbalance is (7).
- (2) **[PROVED]** On short-game values with ambient coherence, every such Brown datum is zero by Theorem 11.
- (3) **[PROVED]** A bare  $C_4$ -central extension is insufficient; a chosen section can encode the phase, but then the framing is supplied data.

Accordingly, under the explicit synchronized four-charge contract, there is no independent mod-8 charge/output problem left. A non-tautological autonomous rule for the nonlinear bit is a generalized **tis**-type problem; on doubled Gold forms it is literally the surviving **tis** problem. Crossing such a readout with the linear bit gives the Brown classes automatically. This closes **over** as a separate invariant/output direction on that contract without promoting the verified  $\sigma$ -charge or finite census to a proof of **tis**, and without asserting that the synchronized pair is already a single canonical normal/misere/loopy outcome semantics. If that single-game internalization is demanded, it is the exact residual boundary and has been transferred rather than proved away.

## References

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